

Baptiste Brument

Ph.D. Student — Computer Vision
IRIT, University of Toulouse

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Education

- 2021 – Present **Ph.D. in Computer Vision: Enhancement of Photographic 3D Reconstruction Techniques**, IRIT, University of Toulouse
Thesis Supervisor: Jean-Denis DUROU / Advisor: Lilian CALVET
- 2020 – 2021 **Master PSMSC - Performance in Software, Multimedia and Scientific Computing**, INP-ENSEEIH, Toulouse
- 2018 – 2021 **Engineering School**, INP-ENSEEIH, Toulouse
Digital Sciences - Image & Multimedia
Competence Areas: Computer Vision — Artificial Intelligence — Geometric and Photometric Methods for 3D Reconstruction
- 2016 – 2018 **Preparatory Classes for Grandes Écoles (PCSI-PSI)**, Lycée Joffre, Montpellier
- 2016 **Scientific Baccalaureate**, Lycée Notre-Dame de la Merci, Montpellier

Professional Experience

- 2021 – Present **Teaching Assistant DCCE**, INP-ENSEEIH, Toulouse
Courses taught: Computer Architecture (BSc), Data Analysis (BSc), Probabilities (BSc), Statistics (BSc), Inverse Problems for 3D (MSc)
- Mar – Sep 2021 **Research Internship**, REVA – IRIT, Toulouse
End of studies project – B. Brument, L. Calvet, J. Mélou, J-D. Durou. 3D Reconstruction of a Convex Polyhedron from its Silhouettes. ORASIS 2021
- Jun – Jul 2020 **Research Internship**, Lab. of Biometry and Bioinformatics, Department of Agricultural and Environmental Biology, University of Tokyo, Japan
Deep-learning based imputation of missing data in historical rice breeding database
- Jun – Jul 2019 **Research Internship**, INRA/CIRAD, Montpellier
Creation of the *openalea.rtfd.io* website containing documentation and interactive tutorials (ReadTheDocs & Jupyter) for the mathematical and computer modeling software OpenAlea
- 2018 **Private Tutor in Physics and Mathematics**, Montpellier

Publications

- Robin Bruneau*, Baptiste Brument*, Yvain Quéau, Jean Mélou, François Bernard Lauze, Jean-Denis Durou, Lilian Calvet, "Multi-view Surface Reconstruction Using Normal and Reflectance Cues," *arXiv preprint arXiv:2506.04115*, 2025.
- Emma Most, Jonas Hein, Frédéric Giraud, Nicola Cavalcanti, Lukas Zingg, Baptiste Brument, Nino Louman, Fabio Carillo, Philipp Fürnstahl, Lilian Calvet, "Acquiring Submillimeter-Accurate Multi-Task Vision Datasets for Computer-Assisted Orthopedic Surgery," in *Proceedings of the International Conference on Information Processing in Computer-Assisted Interventions (IPCAI)*, 2025.

- Antoine Laurent, Benjamin Couprie, Baptiste Brument, Jean Mélou, Yvain Quéau, Carole Fritz, Jean-Denis Durou, "Combining geometric and photometric 3D reconstruction techniques for cultural heritage," *Journal of Cultural Heritage*, vol. 73, pp. 43-51, 2025.
- Benjamin Couprie, Baptiste Brument, Antoine Laurent, Jean Mélou, Yvain Quéau, Jean-Denis Durou, "Assessing the Quality of 3D Reconstruction in the Absence of Ground Truth: Application to a Multimodal Archaeological Dataset," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025.
- Baptiste Brument*, Robin Bruneau*, Yvain Quéau, Jean Mélou, François Bernard Lauze, Jean-Denis Durou, Lilian Calvet, "RNb-NeuS: Reflectance and Normal-based Multi-View 3D Reconstruction," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- Robin Bruneau, Baptiste Brument, Lilian Calvet, Matthew Cassidy, Jean Mélou, Yvain Quéau, Jean-Denis Durou, François Bernard Lauze, "Multi-view stereo of an object immersed in a refractive medium," *Journal of Electronic Imaging (JEI)*, 2024.
- Benjamin Couprie, Jean Mélou, Antoine Laurent, Baptiste Brument, Pierre Gurdjos, Yvain Quéau, Jean-Denis Durou, "Stéréophotométrie avec estimation locale de l'éclairage - Application à la reconstruction 3D du patrimoine archéologique," in *Actes de la conférence Reconnaissance des Formes, Image, Apprentissage et Perception (RFIAP)*, 2024.
- Baptiste Brument, Lilian Calvet, Robin Bruneau, Jean Mélou, Simone Gasparini, Yvain Quéau, François Bernard Lauze, Jean-Denis Durou, "A shape-from-silhouette method for 3D reconstruction of a convex polyhedron," in *Proceedings of SPIE, Sixteenth International Conference on Quality Control by Artificial Vision (QCAV)*, 2023.
- Lilian Calvet, Nicolas Maignan, Baptiste Brument, Silvia Tozza, Jean-Denis Durou, Yvain Quéau, "Multi-view Normal Estimation – Application to Slanted Plane-Sweeping," in *Scale Space and Variational Methods in Computer Vision (SSVM), Lecture Notes in Computer Science*, vol. 14009, Springer, 2023.
- Baptiste Brument, Lilian Calvet, Jean Mélou, Jean-Denis Durou, "Reconstruction 3D d'un polyèdre convexe à partir de ses silhouettes," in *Actes des 18èmes journées francophones des jeunes chercheurs en vision par ordinateur (ORASIS)*, Saint Ferréol, France, 2021.

* **Contribution égale**

Languages

French **Native**
 English **Proficient (C1)**, Cambridge Linguaskill Business: 180+
 Spanish **Basic**
 German **Basic**

Computer Skills

OS **Unix, Windows**
 Languages **Python, Matlab**
 Tools **Latex, Git, Microsoft Office, ReadTheDocs, Jupyter**
 Software **Meshroom, Inkscape, Gimp**